Simple Exponential Smoothing (SES)

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Time Series Analysis

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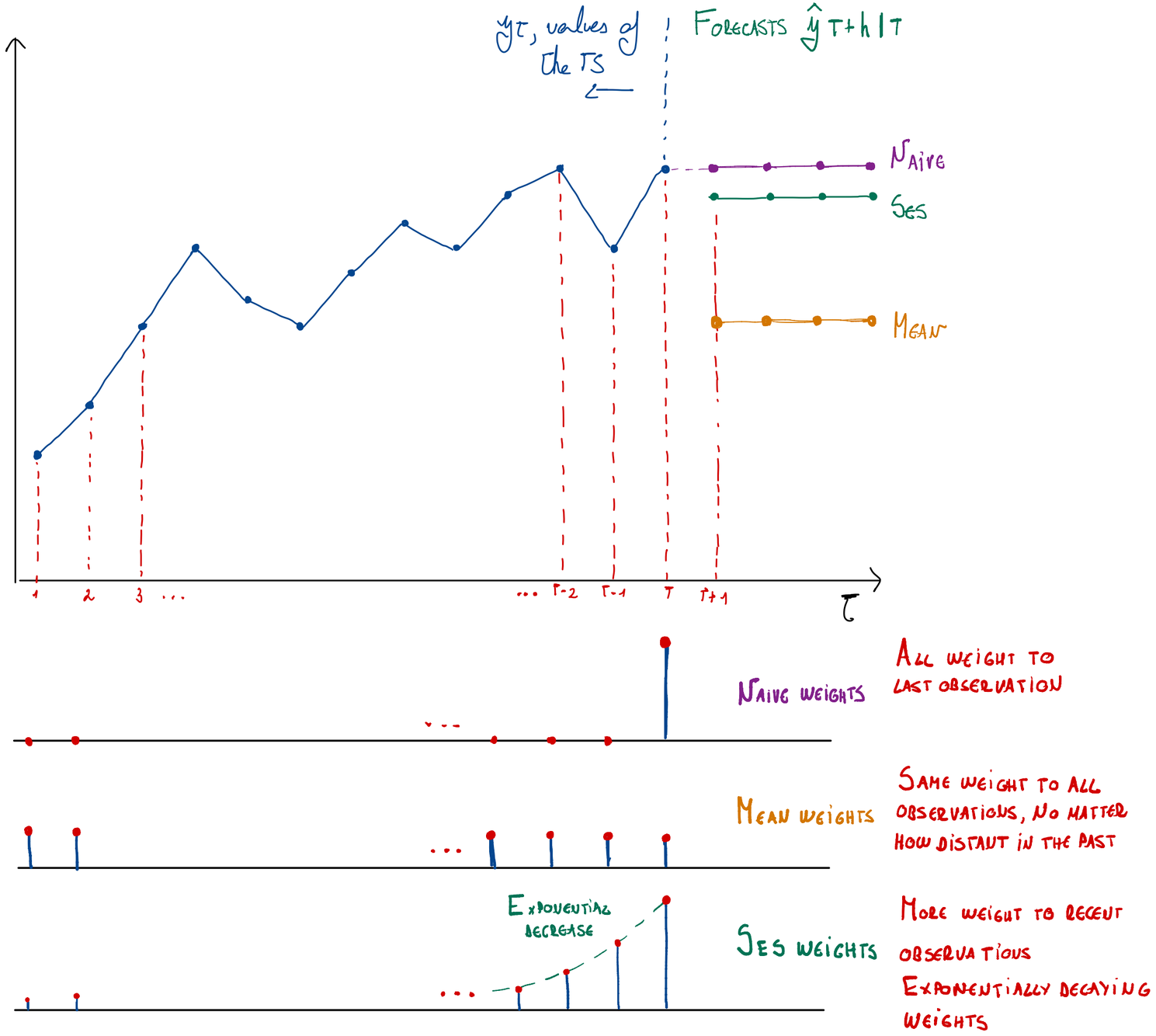

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Geometric sequence (1/3)

$r = \frac{a_n}{a_{n-1}}$ → ratio (given) → Definition

a_1 → given

$a_2 = r \cdot a_1$

$a_3 = r^2 a_1$

$\vdots$

$a_n = r^{n-1} \cdot a_1$

If $r < 1 \Rightarrow \lim_{n \to \infty} a_n = 0$

A specific example:

$a_1 = \alpha \qquad \alpha \in (0, 1]$

$r = (1 - \alpha) \Rightarrow (1 - \alpha) \in [0, 1)$

$a_n = \alpha (1 - \alpha)^{n-1}$

$(1 - \alpha) < 1$

a_n decreases exponentially with $\uparrow n$

$\lim_{n \to \infty} a_n = 0$

Geometric sequence (2/3)

Sum of the first n terms of a geometric sequence

$$S_n = a_1 + \underbrace{a_1 \cdot r}_{a_2} + \underbrace{a_1 \cdot r^2}_{a_3} + \dots + \underbrace{a_1 \cdot r^{n-1}}_{a_n}$$

$$r \cdot S_n = a_1 \cdot r + a_1 \cdot r^2 + a_1 \cdot r^3 + \dots + a_1 \cdot r^n$$

$$S_n - r \cdot S_n = S_n (1 - r) = a_1 - a_1 \cdot r^n = a_1 (1 - r^n)$$

Solving for S_n $\rightarrow$ $S_n = \dfrac{a_1 (1 - r^n)}{(1 - r)}$ $\rightarrow$ $\| r < 1 \Rightarrow \lim_{n \to \infty} S_n = \dfrac{a_1}{1 - r}$

↝ For $r < 1$, S_n converges

Specific case: $\left.\begin{array}{l} a_1 = \alpha \in (0, 1] \\ r = (1 - \alpha) < 1 \end{array}\right\} \rightarrow S_n = \dfrac{\alpha \left(1 - (1 - \alpha)^n\right)}{1 - (1 - \alpha)} = 1 - (1 - \alpha)^n$

$$\left[S_n = \left(1 - (1 - \alpha)^n\right) ; \lim_{n \to \infty} S_n = 1 \right]$$

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Geometric sequence (3/3)

We have shown that, for $\left.\begin{array}{l} a_1 = \alpha \in (0,1] \\ r = (1-\alpha) \in [0,1) \end{array}\right\}$

- $a_n = \alpha (1-\alpha)^{n-1}$; $\lim_{n \to \infty} a_n = 0$ → DISTANT OBSERVATIONS HAVE NEGLIGIBLE WEIGHTS

  ($(1-\alpha) < 1$; $n-1$: EXPONENTIAL DECAY)

- $S_n = 1 - (1-\alpha)^n$; $\lim_{n \to \infty} S_n = 1$ → a_i CAN BE INTERPRETED AS THE PERCENT WEIGHT ASSIGNED TO y_i

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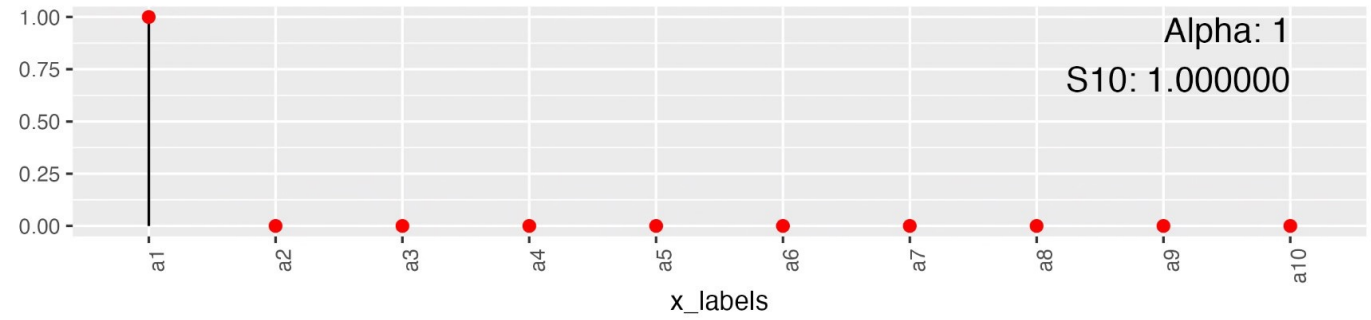
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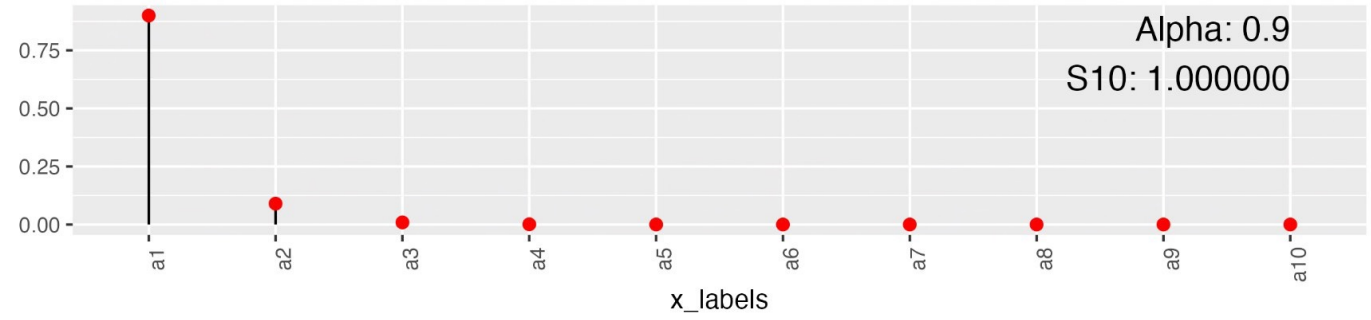
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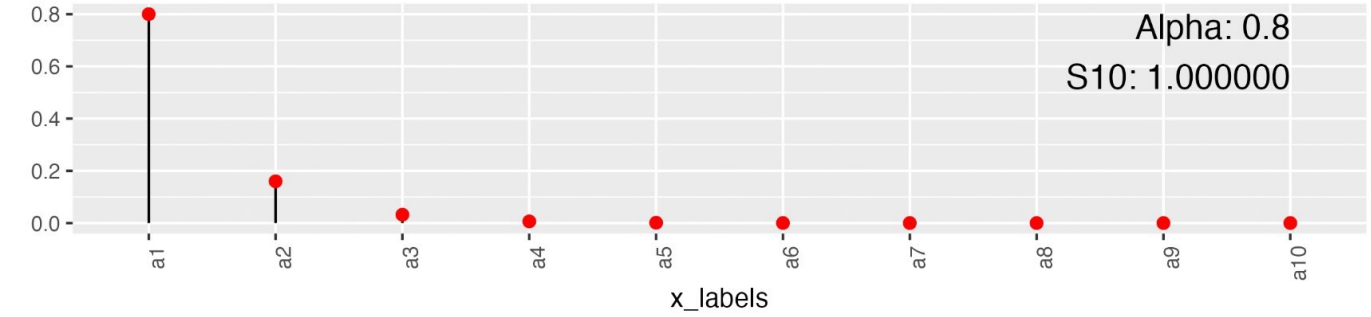
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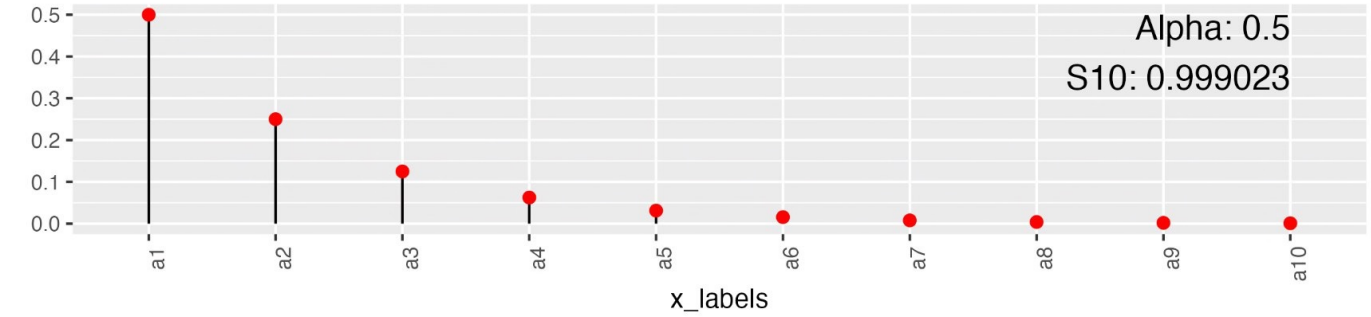
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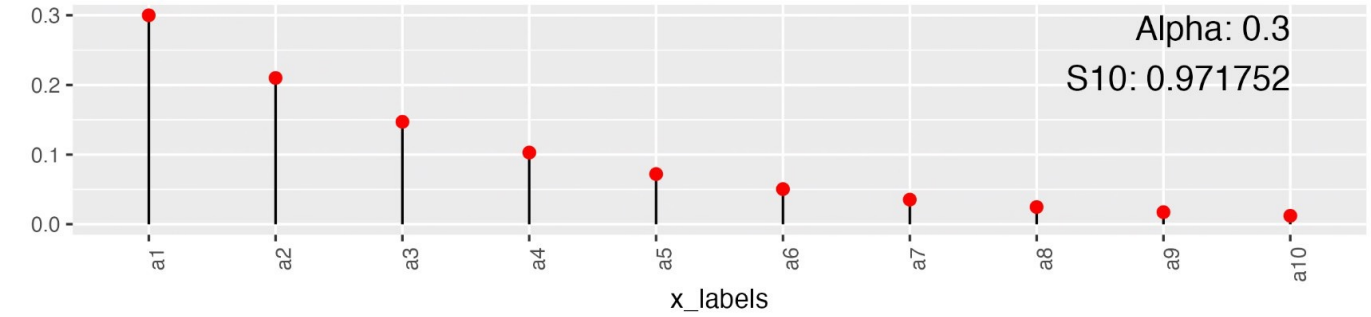
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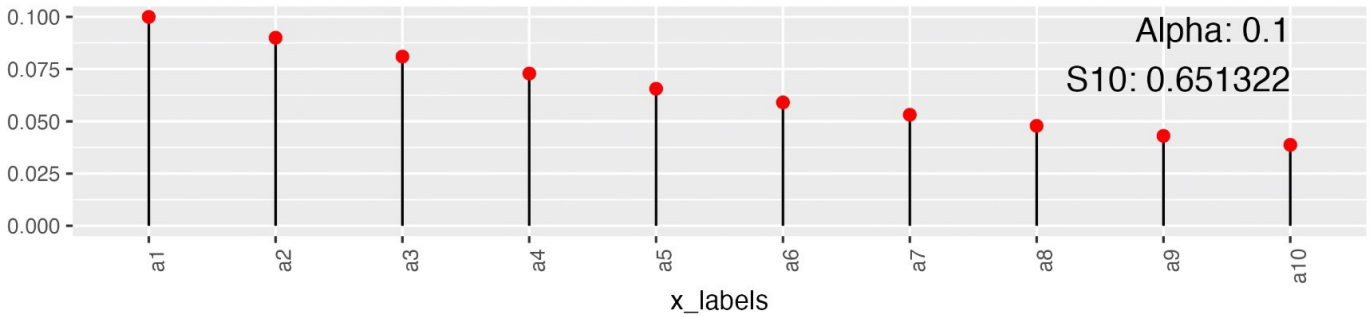
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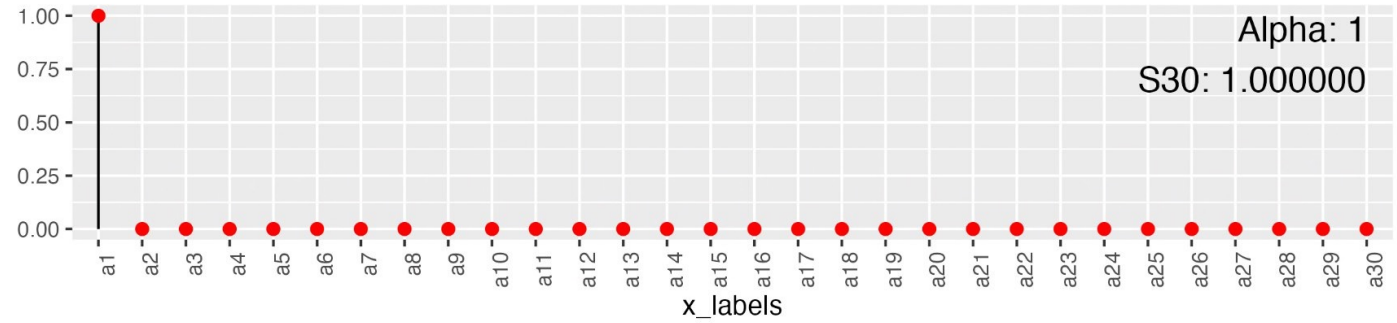
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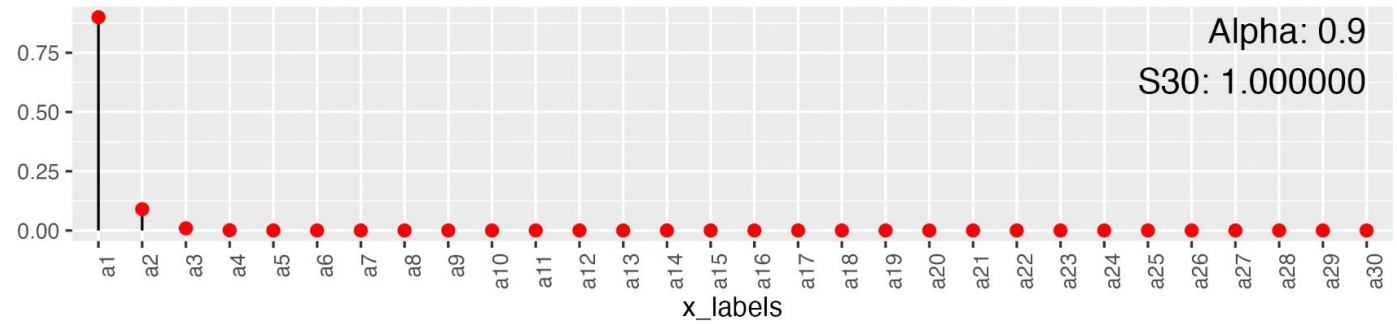
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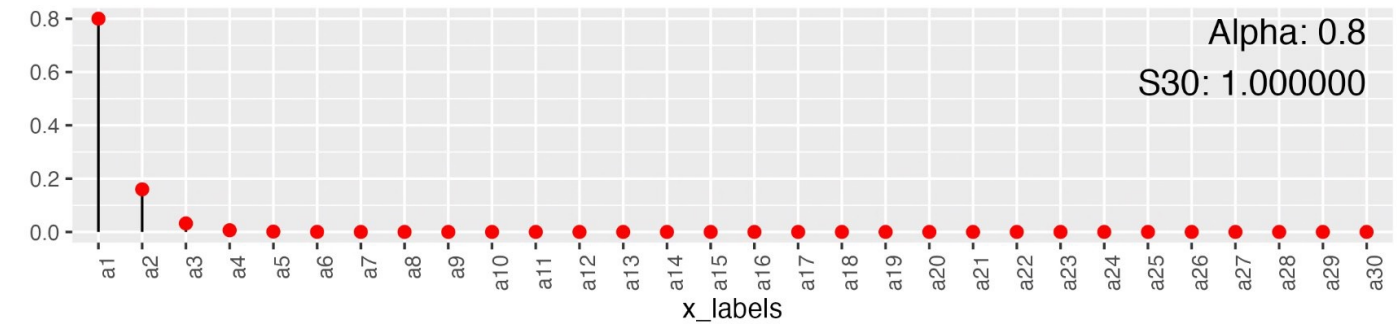
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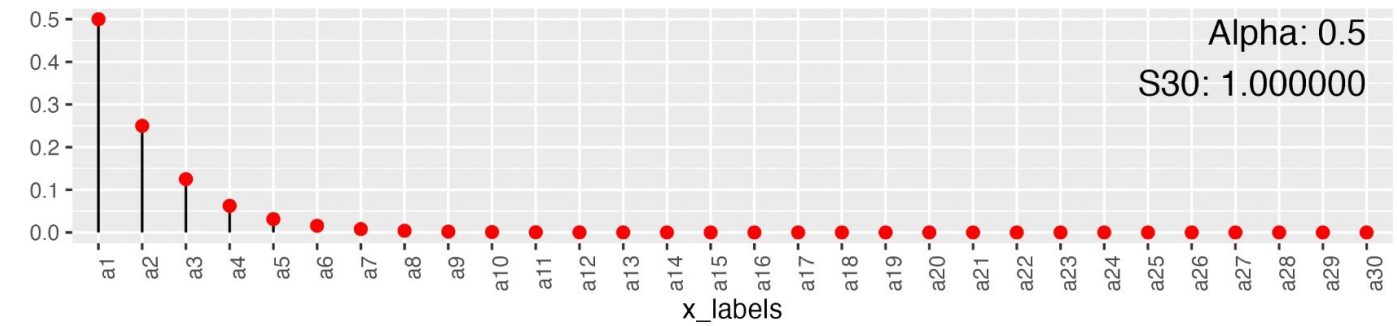
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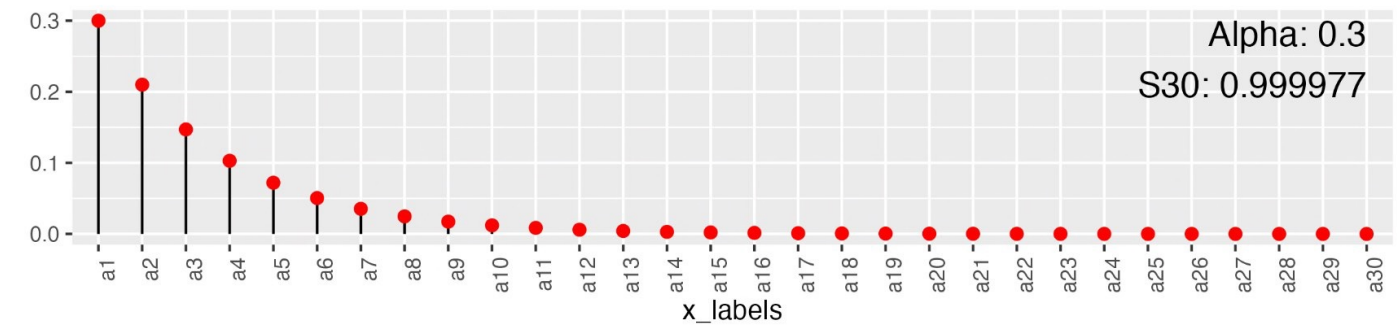
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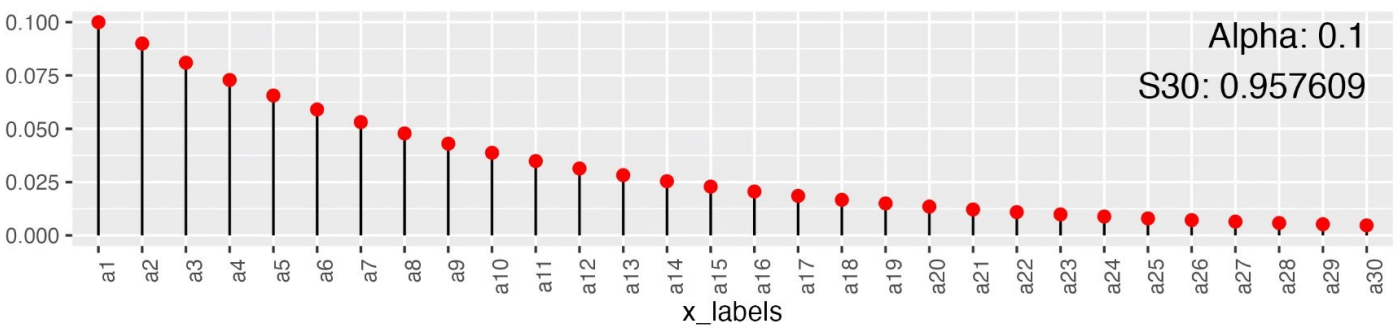
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x_labels

Weighted Average Form

- Given these properties, we may produce a forecast with exp decreasing weights:

$$\hat{y}_{T+1|T} = \underbrace{\alpha}_{a_1} y_T + \underbrace{\alpha(1-\alpha)}_{a_2} y_{T-1} + \underbrace{\alpha(1-\alpha)^2}_{a_3} y_{T-2} + \dots + \underbrace{\alpha(1-\alpha)^{T-1}}_{a_T} y_1$$

$$a_n = \alpha(1-\alpha)^{n-1} \leftarrow \alpha(1-\alpha)^{T-1}$$

$$1 : T-(T-1)$$

- We may express this in a compact manner using the sum operator:

$$\hat{y}_{T+1|T} = \sum_{i=1}^{T} a_i y_{T-(i-1)} = \sum_{i=1}^{T} \alpha(1-\alpha)^{i-1} y_{T-(i-1)} =$$

$$= \sum_{j=0}^{T-1} \alpha(1-\alpha)^{j} y_{T-j}$$

Re-indexing: $j = i-1$

Weighted average form

$$\hat{y}_{T+1|T} = \sum_{j=0}^{T-1} \alpha(1-\alpha)^{j} y_{T-j}$$

By definition of the SES model

$$\hat{y}_{T+h|T} = \hat{y}_{T+1|T}$$

Flat forecasts

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Fitted values using W. Av. Form

Fitted values are DEFINED as $\hat{y}_{t+1|t}$

$$\hat{y}_{T|T-1} = \alpha y_{T-1} + \alpha(1-\alpha) y_{T-2} + \alpha(1-\alpha)^2 y_{T-3} + \dots + \alpha(1-\alpha)^{T-2} y_1$$

$L_1 = T - (T-1)$

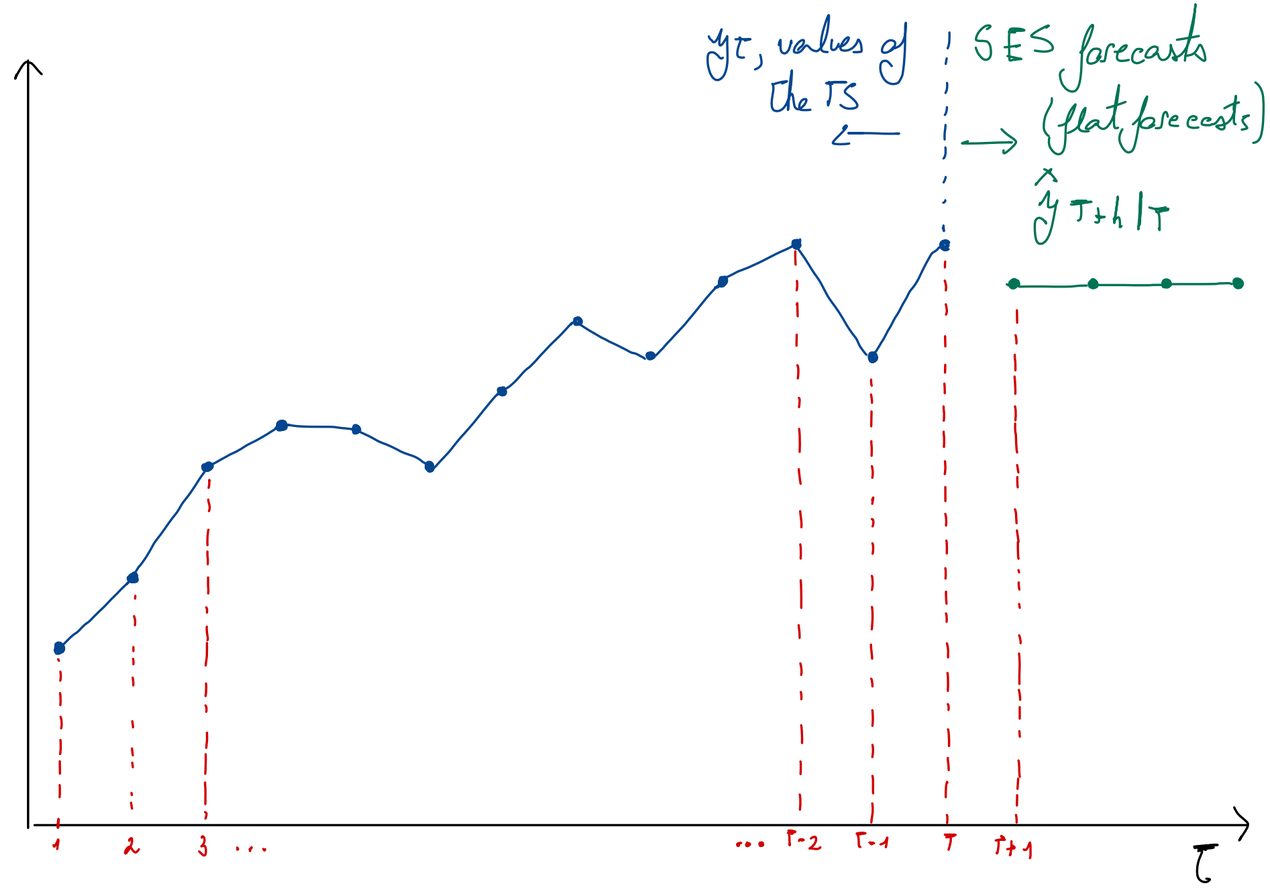
$$\hat{y}_{T-1|T-2} = \alpha y_{T-2} + \alpha(1-\alpha) y_{T-3} + \alpha(1-\alpha)^2 y_{T-4} + \dots + \alpha(1-\alpha)^{T-3} y_1$$

$L_1 = T - (T-1)$

$$\hat{y}_{3|2} = \alpha y_2 + \alpha(1-\alpha) y_1$$

$$\hat{y}_{2|1} = \alpha y_1$$

$$\hat{y}_{1|0} = ? \quad \alpha \cdot y_0 ??$$



PROBLEM:

- → Computationally quite inefficient → The amount of multiplications and sums may be substantially reduced
- → No fitted value at $\tau = 1$
- → Only one parameter α to adjust the fit of our model

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Recursive equation for the fitted values

Weighted average form

$$\hat{y}_{T+1|T} = \sum_{j=0}^{T-1} \alpha(1-\alpha)^j y_{T-j}$$

We are going to use an alternative procedure to compute the fitted values that will solve the problems outlined and enable us to interpret exponential smoothing in terms of "components". This will be most clear in trended and seasonal exponential smoothing.

$$\hat{y}_{T+1|T} = \alpha y_T + \underbrace{\alpha(1-\alpha)^1 y_{T-1} + \alpha(1-\alpha)^2 y_{T-2} + \cdots + \alpha(1-\alpha)^{T-1} y_1}_{(*)}$$

a_1 a_2 a_T

$$\hat{y}_{T|T-1} = \alpha y_{T-1} + \alpha(1-\alpha)^1 y_{T-2} + \alpha(1-\alpha)^2 y_{T-3} + \cdots + \alpha(1-\alpha)^{T-2} y_1$$

$$(1-\alpha)\,\hat{y}_{T|T-1} = \alpha(1-\alpha) y_{T-1} + \alpha(1-\alpha)^2 y_{T-2} + \alpha(1-\alpha)^3 y_{T-3} + \cdots + \alpha(1-\alpha)^{T-1} y_1 = (*)$$

$$\hat{y}_{T+1|T} = \alpha y_T + (*) = \alpha y_T + (1-\alpha)\hat{y}_{T|T-1} \Rightarrow$$

$$\boxed{\hat{y}_{T+1|T} = \alpha y_T + (1-\alpha)\hat{y}_{T|T-1}}$$

Recursive equation for the fitted values

No new info. Direct consequence of the weighted average form

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Recursive computation of fitted values

$$\hat{y}_{T+1|T} = \alpha y_T + (1-\alpha)\hat{y}_{T|T-1}$$

Equation at $\tau = T+1$

c.v.: $\tau = T+1 \Rightarrow T = \tau - 1$

$$\hat{y}_{\tau|\tau-1} = \alpha y_{\tau-1} + (1-\alpha)\hat{y}_{\tau-1|\tau-2}$$

Equation at generic instant τ

Note that $\alpha + (1-\alpha) = 1$

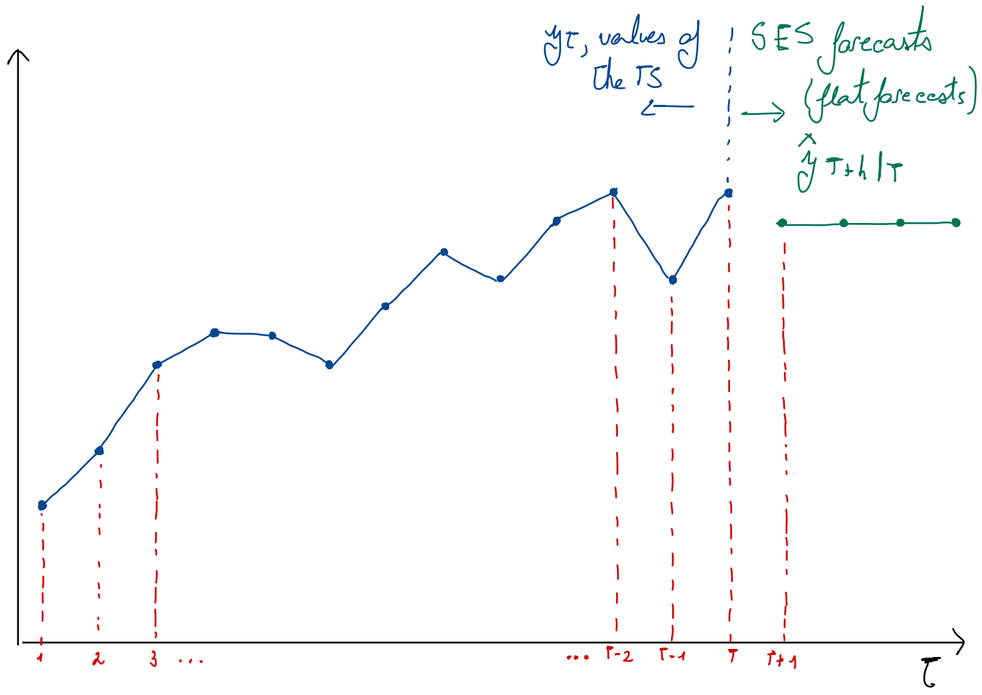
The fitted value at τ is a weighted average of:

- The observation at $\tau-1$ ($y_{\tau-1}$)
- The fitted value at $\tau-1$ ($\hat{y}_{\tau-1|\tau-2}$)

τ	$\hat{y}_{\tau\|\tau-1}$
T	$\hat{y}_{T\|T-1} = \alpha y_{T-1} + (1-\alpha)\hat{y}_{T-1\|T-2}$
$T-1$	$\hat{y}_{T-1\|T-2} = \alpha y_{T-2} + (1-\alpha)\hat{y}_{T-2\|T-3}$
$T-2$	$\hat{y}_{T-2\|T-3} = \alpha y_{T-3} + (1-\alpha)\hat{y}_{T-3\|T-4}$
$\vdots$	$\vdots$
3	$\hat{y}_{3\|2} = \alpha y_2 + (1-\alpha)\hat{y}_{2\|1}$
2	$\hat{y}_{2\|1} = \alpha y_1 + (1-\alpha)\hat{y}_{1\|0}$
1	$\hat{y}_{1\|0} = \ell_0$

ℓ_0: Initial value for the level

↳ New parameter



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Fitted Values as updating approximation

Note that $\alpha + (1-\alpha) = 1$

$$\hat{y}_{\tau|\tau-1} = \alpha\, y_{\tau-1} + (1-\alpha)\,\hat{y}_{\tau-1|\tau-2} \quad \oplus \quad \hat{y}_{1|0} = l_0$$

Equation at generic instant τ

The fitted value at τ $\hat{y}_{\tau|\tau-1}$ is a weighted average of:

- → The observation at $\tau-1$ $y_{\tau-1}$
- → The fitted value at $\tau-1$ $\hat{y}_{\tau-1|\tau-2}$

→ Getting a new observation at τ y_τ allows us to update our approximation for the next point in time $\tau+1$

$$\hat{y}_{\tau+1|\tau} = \alpha \cdot y_\tau + (1-\alpha)\,\hat{y}_{\tau|\tau-1}$$

New approx | New info (observation) | Previous approx

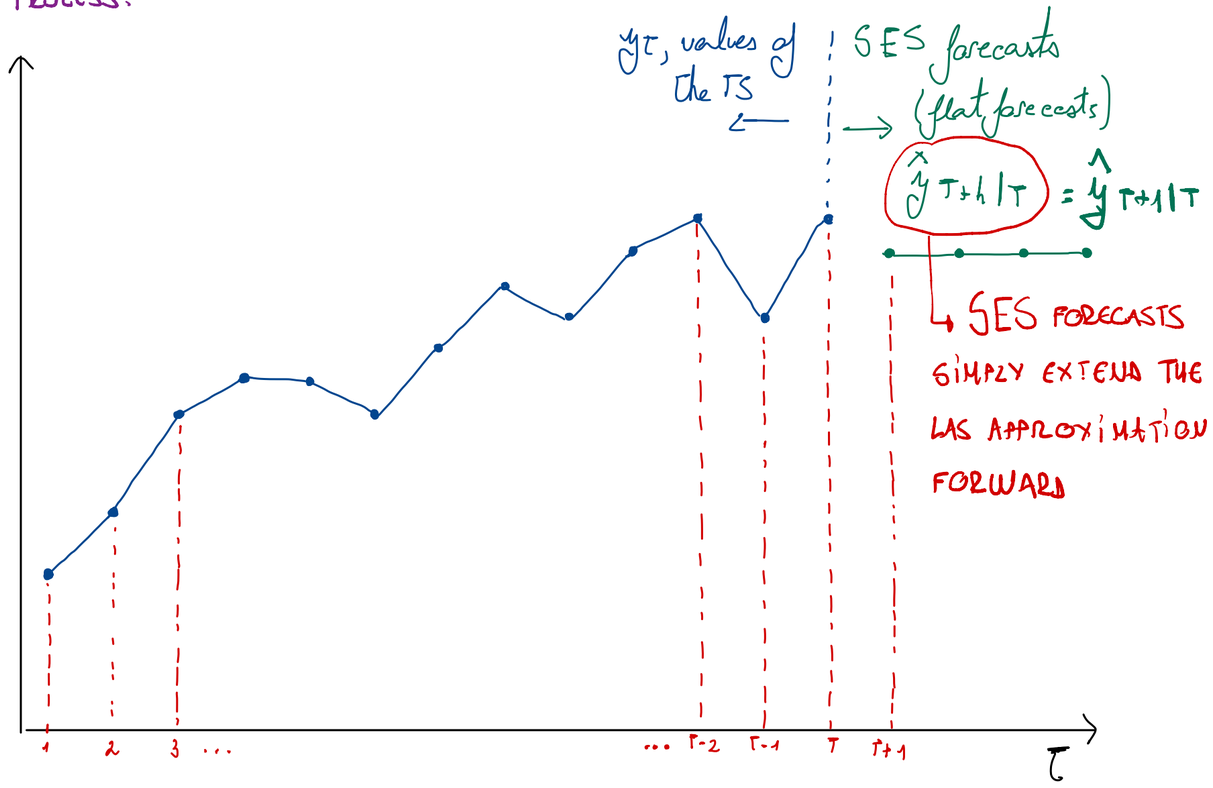
→ Initial level. Guess. Parameter of process.

τ	$\hat{y}_{\tau\|\tau-1}$
1	$\hat{y}_{1\|0} = l_0$
2	$\hat{y}_{2\|1} = \alpha\, y_1 + (1-\alpha)\,\hat{y}_{1\|0}$
3	$\hat{y}_{3\|2} = \alpha \cdot y_2 + (1-\alpha)\,\hat{y}_{2\|1}$
⋮	⋮
$\tau-2$	$\hat{y}_{\tau-2\|\tau-1} = \alpha \cdot y_{\tau-1} + (1-\alpha)\,\hat{y}_{\tau-3\|\tau-4}$
$\tau-1$	$\hat{y}_{\tau-1\|\tau-2} = \alpha \cdot y_{\tau-2} + (1-\alpha)\,\hat{y}_{\tau-2\|\tau-1}$
τ	$\hat{y}_{\tau\|\tau-1} = \alpha \cdot y_{\tau-1} + (1-\alpha)\,\hat{y}_{\tau-1\|\tau-2}$

Updated Approximation

Observation: new info

Previous Approximation



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FITTING THE MODEL AND GETTING FORECASTS

Given α and l_0, the equations below provide the fitted values at each point:

	t	$\hat{y}_{t\|t-1}$
TRAINING	1	$\hat{y}_{1\|0} = l_0$
	2	$\hat{y}_{2\|1} = \alpha y_1 + (1-\alpha)\hat{y}_{1\|0}$
	3	$\hat{y}_{3\|2} = \alpha y_2 + (1-\alpha)\hat{y}_{2\|1}$
	⋮	⋮
	$T-2$	$\hat{y}_{T-2\|T-1} = \alpha y_{T-1} + (1-\alpha)\hat{y}_{T-3\|T-4}$
	$T-1$	$\hat{y}_{T-1\|T-2} = \alpha y_{T-2} + (1-\alpha)\hat{y}_{T-2\|T-1}$
	T	$\hat{y}_{T\|T-1} = \alpha y_{T-1} + (1-\alpha)\hat{y}_{T-1\|T-2}$
TEST	$T+1$	$\hat{y}_{T+1\|T} = \alpha y_T + (1-\alpha)\hat{y}_{T\|T-1}$

- $\hat{y}_{1|0}$: Initial level. Guess. Parameter of process.
- $\hat{y}_{2|1}$: Updated Approximation
- y_1: Observation: new info
- $\hat{y}_{1|0}$: Previous Approximation

$l = \hat{y}_{T+h|T}$ (SES has flat forecasts)

1. Determine initial values

l_0, α

2. Apply the component form of the equations to calculate the fitted values.
That is, at each point in time t compute:

- $\hat{y}_{1|0} = l_0$
- $\hat{y}_{2|1} = \alpha\, y_1 + (1-\alpha)\, \hat{y}_{1|0}$
- $\hat{y}_{3|2} = \alpha\, y_2 + (1-\alpha)\, \hat{y}_{2|1}$
- ⋮
- $\hat{y}_{T|T-1} = \alpha\, y_{T-1} + (1-\alpha)\, \hat{y}_{T-1|T}$

EXCEL FILE

3. Compute the Sum of Square Residuals

$$\text{SSE} = \sum_{t=1}^{T}(y_t - \hat{y}_{t|t-1})^2 = \sum_{t=1}^{T} e_t^2$$

4. Minimality of SSE satisfied?

NO → 1. Determine initial values

YES → Model has been fitted.

Flat forecasts:

$$\hat{y}_{T+h|T} = \hat{y}_{T+1|T} = \alpha\, y_T + (1-\alpha)\, \hat{y}_{T|T-1}$$

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PARAMETERS: α AND l_0

SMOOTHING PARAMETER

- α: determines the weight assigned to the new info ($\alpha \cdot y_\tau$) and the previous approximation ($(1-\alpha)\,\hat{y}_{\tau|\tau-1}$)
  - EQUIVALENT TO: determines how the weight assigned to past observations decays
- l_0: influences the value of the fitted exponential smoothing
  - Its influence becomes weaker the more distant from $\tau = 1$ the process gets:

REMINDER
$\alpha \in (0,1] \rightarrow 1-\alpha \in [0,1)$
$\alpha + (1-\alpha) = 1 \rightarrow$ Weighted average

$\hat{y}_{1|0} = l_0$

$\tau = 1$: $\hat{y}_{1|0} = l_0$

$\tau = 2$: $\hat{y}_{2|1} = \alpha y_1 + (1-\alpha)\,\hat{y}_{1|0} = \alpha y_1 + (1-\alpha)\,l_0$

$\tau = 3$: $\hat{y}_{3|2} = \alpha y_2 + (1-\alpha)\,\hat{y}_{2|1} = \alpha y_2 + (1-\alpha)\left[\alpha y_1 + (1-\alpha) l_0\right] =$

$= \alpha y_2 + \alpha(1-\alpha) y_1 + \underbrace{(1-\alpha)^2}_{<1} l_0$

$\vdots$

GENERIC POINT τ: $\hat{y}_{\tau|\tau-1} = \alpha y_{\tau-1} + \alpha(1-\alpha) y_{\tau-2} + \alpha(1-\alpha)^2 y_{\tau-3} + \dots + \alpha(1-\alpha)^{\tau-2} y_1 + (1-\alpha)^{\tau-1} \cdot l_0$

WEIGHTED AVERAGE FORM

$1 = \tau - (\tau - 1)$

INFLUENCE OF l_0 DECAYS EXPONENTIALLY WITH INCREASING τ

$(1-\alpha)^{\tau-1} < 1$

Definition of the level l_t

We define the level at t: $\boxed{l_t = \hat{y}_{t+1|t}} \Rightarrow l_0 = \hat{y}_{1|0}$

t	$\hat{y}_{t\|t-1}$	
T	$\hat{y}_{T\|T-1} = \alpha y_{T-1} + (1-\alpha)\hat{y}_{T-1\|T-2}$	l_{T-1}, l_{T-2}
$T-1$	$\hat{y}_{T-1\|T-2} = \alpha y_{T-2} + (1-\alpha)\hat{y}_{T-2\|T-3}$	l_{T-2}, l_{T-3}
$T-2$		
$\vdots$	$\vdots$	$\vdots$
3	$\hat{y}_{3\|2} = \alpha y_2 + (1-\alpha)\hat{y}_{2\|1}$	l_2, l_1
2	$\hat{y}_{2\|1} = \alpha y_1 + (1-\alpha)\hat{y}_{1\|0}$	l_1, l_0
1	$\hat{y}_{1\|0} = l_0$	Initial value for the level

In summary, with this definition of l_t

$$\hat{y}_{t+1|t} = \alpha \cdot y_t + (1-\alpha)\hat{y}_{t|t-1}$$

$$\boxed{l_t = \alpha \cdot y_t + (1-\alpha) l_{t-1}}$$

SMOOTHING EQUATION

(ALTERNATIVE FORM OF RECURSIVE EQUATION FOR FITTED VALUES)

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Component form of the SES equations

Forecast equation $\hat{y}_{T+h|T} = \hat{y}_{T+1|T} = \ell_T$ (flat forecasts; definition of level)

$\ell_t = \hat{y}_{t+1|t}$

$\ell_t = \hat{y}_{t+1|t}$ (definition of level)

Smoothing equation: $\hat{y}_{t+1|t} = \alpha y_t + (1-\alpha)\hat{y}_{t|t-1}$ ($\hat{y}_{t+1|t} = \ell_t$, $\hat{y}_{t|t-1} = \ell_{t-1}$)

$\leftrightarrow \ell_t = \alpha y_t + (1-\alpha)\ell_{t-1}$

SES - Equations (2 equivalent forms)

Weighted average form

$$\hat{y}_{T+1|T} = \sum_{j=0}^{T-1} \alpha(1-\alpha)^j y_{T-j}$$

Component form

Forecast eq: $\hat{y}_{T+h|T} = \hat{y}_{T+1|T} = \ell_T$

Smoothing eq: $\ell_t = \alpha y_t + (1-\alpha)\ell_{t-1}$

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Exponential Moving Average and ℓ_t

Investopedia:

Formula for Exponential Moving Average (EMA)

$$EMA_{\text{Today}} = \left(\text{Value}_{\text{Today}} * \left(\frac{\text{Smoothing}}{1+\text{Days}}\right)\right) + EMA_{\text{Yesterday}} * \left(1 - \left(\frac{\text{Smoothing}}{1+\text{Days}}\right)\right)$$

α — Usually ~ 2

e.g.: 20 days → EMA_{20}

α

where:

EMA = Exponential moving average

$$EMA_{\text{TODAY}} = \alpha \cdot y_{\text{TODAY}} + (1-\alpha)\, EMA_{\text{YESTERDAY}}$$

$$EMA_t = \alpha \cdot y_t + (1-\alpha)\, EMA_{t-1}$$

Component form of SES

Forecast. eq: $\hat{y}_{T+h|T} = \hat{y}_{T+1|T} = \ell_T$

Smoothing. eq: $\ell_t = \alpha \cdot y_t + (1-\alpha)\,\ell_{t-1}$

→ The level is the EMA of the time series

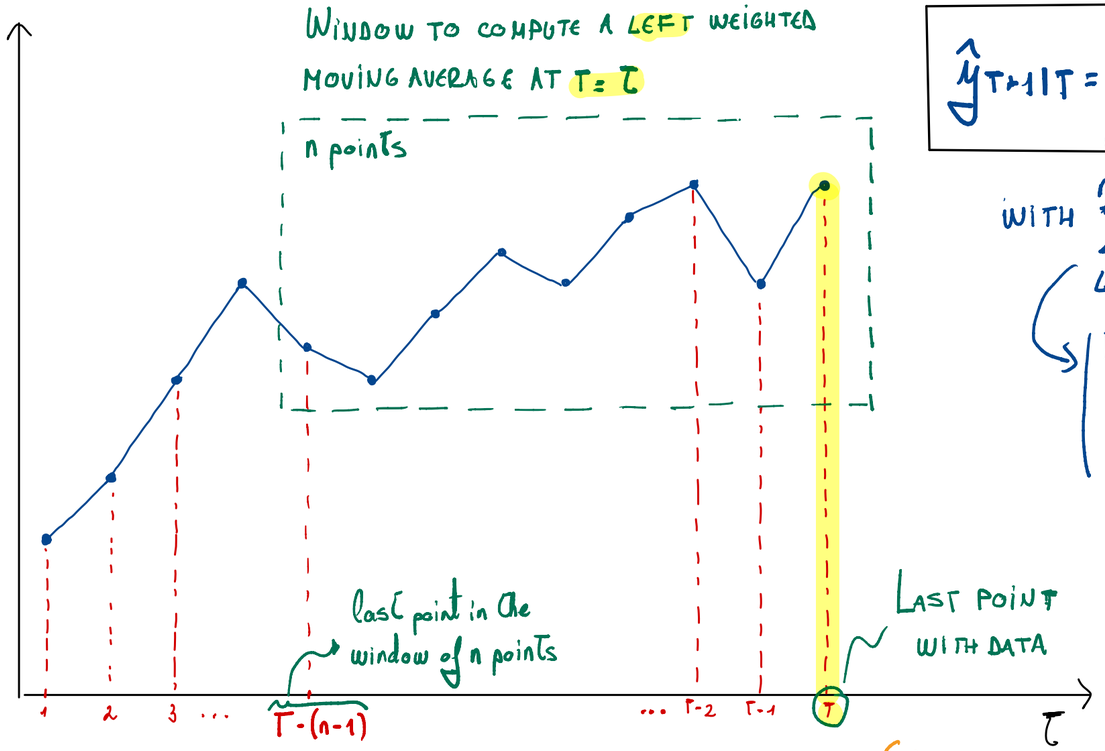
EMA_0 required (ℓ_0)

Simple Moving Average (next slide)

Calculating the EMA

Calculating the EMA requires one more observation than the SMA. Suppose that you want to use 20 days as the number of observations for the EMA. Then, you must wait until the 20th day to obtain the SMA. On the 21st day, you can then use the SMA from the previous day as the first EMA for yesterday.

Exponential vs Simple Moving Average



Left weighted moving average of n points

$$\hat{y}_{T+1|T} = \sum_{i=0}^{n-1} w_i \cdot y_{T-i}$$

with $\sum_{i=0}^{n-1} w_i = 1$

↳ Each w_i is the percent contribution of y_i

- **Simple (Left) Moving Average**

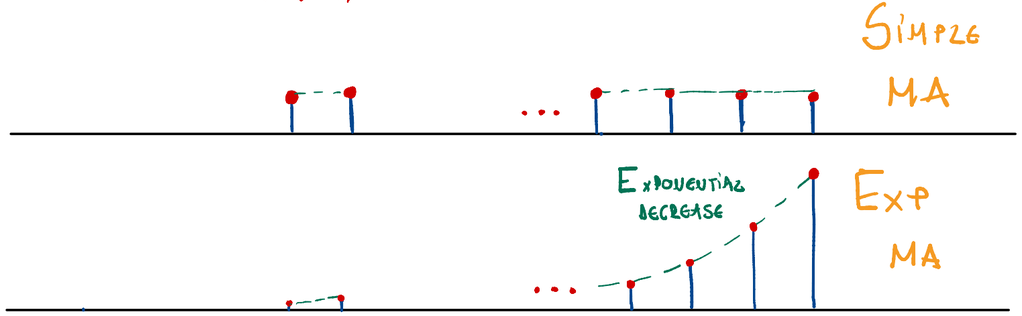
$$w_i = \frac{\sum_{j=0}^{n-1} y_j}{n} \quad \forall_i$$

↳ Same weight for all observations

- **Exponential Moving Average**

$$w_i = \alpha(1-\alpha)^i$$

↳ Obtained from geometric as in SES.



Same weight to all observations, no matter how distant in the past

More weight to recent observations

Exponentially decaying weights

Note: if all points from 1 to T are included in the window ($n = T$)

SMA → Mean Model

EMA → Simple Exponential Smoothing.

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